

HingeMem: Boundary Guided Long-Term Memory with Query Adaptive Retrieval for Scalable Dialogues

Yijie Zhong
zhongyj@tongji.edu.cn
College of Design and Innovation,
Tongji University
Shanghai, China

Yunfan Gao
gaoyunfan1602@gmail.com
Shanghai Research Institute for
Intelligent Autonomous Systems,
Tongji University
Shanghai, China

Haofen Wang*
carter.whfcarter@gmail.com
College of Design and Innovation,
Tongji University
Shanghai, China

Abstract

Long-term memory is critical for dialogue systems that support continuous, sustainable, and personalized interactions. However, existing methods rely on continuous summarization or OpenIE-based graph construction paired with fixed Top- k retrieval, leading to limited adaptability across query categories and high computational overhead. In this paper, we propose HingeMem, a boundary-guided long-term memory that operationalizes event segmentation theory to build an interpretable indexing interface via boundary-triggered hyperedges over four elements: person, time, location, and topic. When any such element changes, HingeMem draws a boundary and writes the current segment, thereby reducing redundant operations and preserving salient context. To enable robust and efficient retrieval under diverse information needs, HingeMem introduces query-adaptive retrieval mechanisms that jointly decide (a) *what to retrieve*: determine the query-conditioned routing over the element-indexed memory; (b) *how much to retrieve*: control the retrieval depth based on the estimated query type. Extensive experiments across LLM scales (from 0.6B to production-tier models; e.g., Qwen3-0.6B to Qwen-Flash) on LOCOMO show that HingeMem achieves approximately 20% relative improvement over strong baselines without query categories specification, while reducing computational cost (68%↓ question answering token cost compared to HippoRAG2). Beyond advancing memory modeling, HingeMem's adaptive retrieval makes it a strong fit for web applications requiring efficient and trustworthy memory over extended interactions.

CCS Concepts

• Computing methodologies → Natural language generation.

Keywords

Dialogue System; Personalized Memory; Hippocampal-Cortical Interaction; Query Adaptive Retrieval

ACM Reference Format:

Yijie Zhong, Yunfan Gao, and Haofen Wang. 2026. HingeMem: Boundary Guided Long-Term Memory with Query Adaptive Retrieval for Scalable Dialogues. In *Proceedings of the ACM Web Conference 2026 (WWW '26)*.

*Corresponding author.



This work is licensed under a Creative Commons Attribution 4.0 International License. WWW '26, Dubai, United Arab Emirates.

© 2026 Copyright held by the owner/author(s).
ACM ISBN 979-8-4007-2307-0/2026/04
<https://doi.org/10.1145/3774904.3792089>

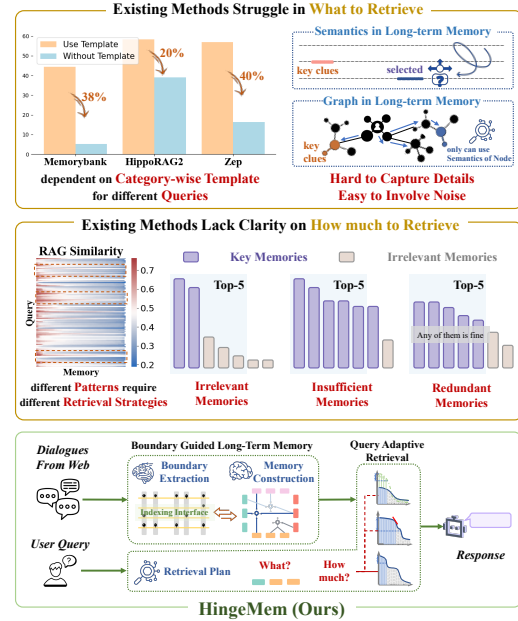


Figure 1: Existing methods face two challenges when facing various queries: they struggle to identify key clues and find it difficult to control the retrieval process. HingeMem simulates the human cortex and hippocampus to construct boundary-guided long-term memory, employing query-adaptive retrieval to generate unique retrieval plans.

April 13–17, 2026, Dubai, United Arab Emirates. ACM, New York, NY, USA, 12 pages. <https://doi.org/10.1145/3774904.3792089>

1 Introduction

As Large Language Models (LLMs) advance rapidly, dialogue systems increasingly power everyday assistants and long-lived companions [5, 14]. In human-human communication, people rely on memory to retain and leverage past information [47]. Intelligent Web applications built on dialogue systems similarly require robust long-term memory to support continual and personalized interactions [42]. This paper focuses on advancing long-term memory modeling for dialogue systems.

Recent studies have explored long-term memory from multiple perspectives [1, 37, 54], including lightweight plugins [8, 33], centralized services [7, 34], and system-level architectures [25]. These

approaches typically follow two directions: (i) continuously recording, summarizing, and updating dialogue histories as memory [2, 23, 39, 44], or (ii) transforming conversations into structured graph-based memory using techniques such as OpenIE [19, 40, 57, 58]. However, both directions often overlook details of event dynamics in the dialogue history and struggle to flexibly control retrieval across diverse queries. When faced with diverse queries without explicit category specification, responses are unstable and performance degrades substantially. Moreover, memory construction and maintenance incur high computational costs without commensurate gains¹, which limits practical deployment in Web applications.

Continuous memory writing and unconstrained OpenIE processing frequently result in incomplete detail capture and substantial overhead. Existing methods rely on semantic similarity-based retrieval over plain text or graph nodes, which hinders the precise alignment of query information and key evidence for various queries. As illustrated in Figure 1, performance drops around 30% when query categories are unspecified. Moreover, most methods employ fixed Top- k retrieval strategies (e.g. Memorybank [56] Top-5, Mem0 [7] Top-30), which may introduce both redundancy and noise. The effectiveness of such fixed- k quickly saturates or even declines for different categories of queries [49]. This effect becomes especially evident in real dialogue scenarios. Similarity patterns in the heatmap of vanilla RAG in Figure 1 indicate that fixed- k retrieval is suboptimal. Some queries require retrieving nearly all relevant segments (e.g. ‘How many meetings have we held in total?’ and ‘What sports do I enjoy?’). Whereas others are best addressed with only a few or any one of the relevant segments (e.g. ‘What was I doing at 12:00 yesterday?’ and ‘Have I dined with Alex before?’). Forcing the retrieval of low-relevance content or discarding high-relevance content leads to decreases in accuracy and efficiency.

In summary, accurate and efficient long-term memory must address two complementary challenges: 1) **What to retrieve**: design a general, interpretable, and composable indexing interface that aligns the query with the right evidence elements and their combinations. 2) **How much to retrieve**: determine retrieval depth adaptively, matching the evidence size required by different queries.

Inspired by cognitive neuroscience, we propose HingeMem, a boundary-guided long-term memory with query-adaptive retrieval for dialogue systems. Drawing on event segmentation theory, HingeMem exposes an interpretable and generalizable indexing interface over four key elements: person, time, location, and topic. When any of these elements changes, HingeMem writes a boundary-aligned segment as a structured hyperedge, reducing redundant operations while preserving salient details. These hyperedges connect element-specific nodes to concise descriptions, yielding indices that support both semantic and graph-informed retrieval. During retrieval, HingeMem generates query-adaptive retrieval plans that jointly decide what and how much to retrieve. The plan identifies element constraints and priorities, re-ranks candidate hyperedges, and applies adaptive stopping policies. We categorize queries into three retrieval-oriented types: recall-priority, precision-priority, and judgment, each with tailored stopping criteria. Experiments across LLM scales show that HingeMem achieves superior performance with reduced computational cost, and it operates without

category-specific question templates, making it adaptable to diverse scenarios. The contributions are as follows:

- We introduce HingeMem, a neuro-inspired, boundary-guided memory with query-adaptive retrieval. It writes memory segments at element changes, minimizing continuous writing and providing explicit, interpretable retrieval interfaces.
- We propose a query-adaptive retrieval mechanism that optimizes what to and how much to retrieve, overcoming the limitations of fixed-depth retrieval and reducing computational cost.
- Experiments show consistent gains on LOCOMO without relying on category-specific templates and show robust effectiveness across LLM scales, highlighting HingeMem’s practicality for Web applications requiring efficient long-term memory.

2 Related Work

2.1 Long-term Memory Mechanism

Prior work on long-term memory for dialogue systems aims to explicitly retain and retrieve historical dialogues and user profiles [47]. Most methods instantiate memory as either unstructured plain text or structured graphs. On the text side, SCM [43] introduces a Self-Controlled Memory framework with a memory stream and controller. TiM [26] distills the ‘thoughts’ in raw dialogues to support recalling and post-thinking. A line of work [27, 44] employs different summarization strategies to construct unstructured memories. However, purely textual memory often struggles to preserve fine-grained details and lacks explicit indices for targeted retrieval. On the graph side, following GraphRAG [11], several studies exploit graphs to build structured memories. EMG-RAG [45] maintains an editable personal knowledge graph to support insertion and deletion of memories. Several efforts leverage neuro-symbolic techniques to build query-specific retrieval paths [15, 31], but these approaches predominantly target temporal questions and are computationally expensive. While graphs provide better structural information than plain text, unconstrained extraction may introduce noise, making query alignment non-trivial for various queries.

Beyond engineering choices, researchers have drawn on insights from cognitive neuroscience [12]. Memorybank [56] leverages the Ebbinghaus forgetting curve to guide memory updates. A representative example is HippoRAG [18, 19], which operationalizes hippocampal indexing theory by treating the LLM as an artificial ‘neocortex’ and maintaining a schema-free OpenIE knowledge graph as a ‘hippocampal-like index’. This architecture is practical, yet it relies on a single unified index for the entire events with fixed Top- k retrieval. In contrast, we draw on event segmentation theory and hippocampal–cortical interactions. We encode memory at event boundaries and construct hyperedges. Versus text-based systems, our boundary-triggered hyperedges preserve fine-grained details and expose explicit indices. Versus OpenIE-centric graph memories, our element-indexed hyperedges improve interpretability and query alignment. Versus fixed-depth retrieval, we employ query-adaptive stopping tied to retrieval-oriented query types.

2.2 Long-term Memory Evaluation

For many years, researchers have developed datasets through various methods, including crowd-sourcing, web data collection, and

¹LangMem incurs a token overhead exceeding 1000× the total tokens in all dialogues

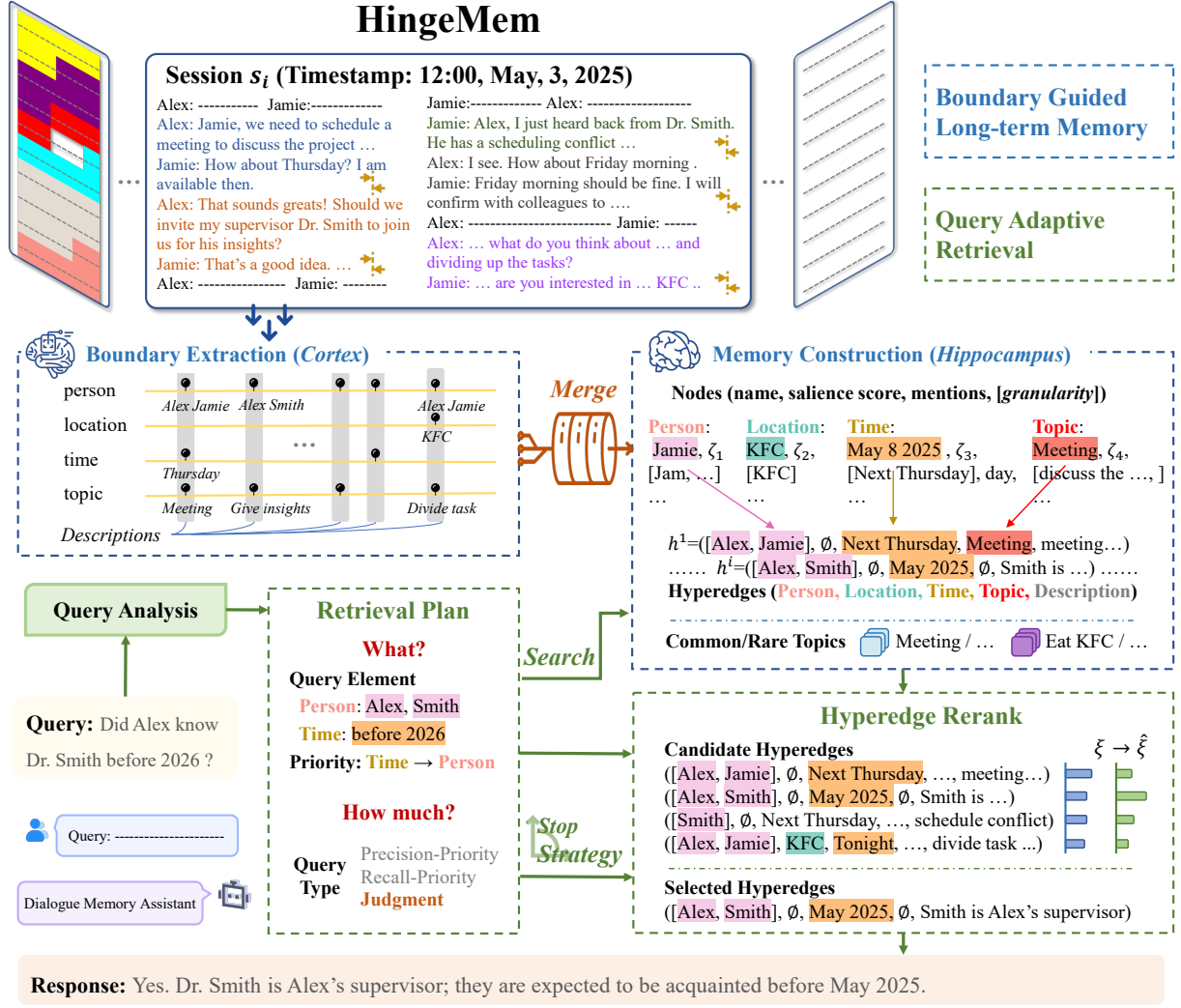


Figure 2: Overall Boundary-guided Long Term Memory construction and Query Adaptive Retrieval process of HingeMem.

generation [9, 47], to facilitate the evaluation and technical iteration of long-term memory for dialogue systems. MSC [48] and Conversation Chronicles [21] serve as a foundational resource for early research. However, they are constrained by the conversation length and the correlation between dialogues. DuLeMon [50] focuses on long-term persona memory dialogues in Chinese contexts but is limited by its lack of scenario diversity. TimelineQA [41] concentrates exclusively on time series reasoning, neglecting other memory elements. Memorybank [56] and PerLTQA [10] push forward comprehensive evaluations. Memorybank is limited in scale and lacks annotations, and PerLTQA's scenarios are relatively uniform. Recent benchmarks broaden the perspective: MADial-Bench [20] evaluates how long-term memory supports user emotions, while LAMP [36] emphasizes whether responses can mimic the user's personality and behavior. SHARE [22] and HiCUPID [30] explore long-term dialogues but face singular memory source and narrow question category constraints.

LongMemEval [46] and LOCOMO [29] are most related to our work. LongMemEval establishes a benchmark for evaluating long-term interactive memory. However, each dialogue contains only a single question, making a comprehensive assessment inefficient. In contrast, LOCOMO focuses on ultra-long dialogue memory with a data design closely aligned with realistic multi-turn scenarios, offering advantages in memory span and question diversity. Accordingly, we adopt LOCOMO and follow its category-wise F1 score and an additional LLM-as-a-Judge setting for comparability.

3 HingeMem

3.1 Overview

Figure 2 presents our proposed neuro-inspired HingeMem, comprising two key components: (a) Boundary-Guided Long-Term Memory, which leverages node indices and hyperedges; and (b) Query-Adaptive Retrieval, which features retrieval planning, hyperedge reranking, and adaptive stopping. A simulated *cortex* performs

dialogue boundary extraction within each session, extracts key elements from segments, and organizes them into a hyperedge. A simulated *hippocampus* then consolidates the resulting hyperedges and element-indexed nodes into long-term memory. During retrieval, HingeMem produces a retrieval plan for each query, explicitly specifying both *what* to retrieve and *how much* to retrieve.

3.2 Boundary Guided Long-Term Memory

Cognitive neuroscience indicates that the brain does not encode information uniformly. Instead, it privileges event boundaries [3, 16]. At such transitions, activity in the prefrontal cortex (PFC) intensifies, hippocampal activation increases, and interactions between the hippocampus and PFC, collectively facilitating memory formation [35, 55]. Converging evidence further shows that indexing at event offsets or boundaries is beneficial, whereas mid-event indexing can be counterproductive [13, 28].

Event Segmentation Theory (EST) [52] holds that segmentation is automatic [53], operates at multiple scales [6], and is triggered by salient changes in time, space, objects or characters, goals, or causality [6]. Such boundaries partition continuous streams, reducing representational load and supporting memory. Critically, boundary-aligned information is more easily recalled, and temporal order is more reliably preserved. Recent fMRI work further underscores the importance of “encoding at boundaries,” showing that hippocampal–cortical interactions at these points promote persistent and detailed memory representations [4]. Therefore, we design boundary-centric encoding in HingeMem by coupling a cortex-like boundary extractor with a hippocampus-like memory constructor.

3.2.1 Dialogue Boundary Extraction. Inspired by Event Segmentation Theory, we identify boundaries in dialogues based on changes in key elements. In dialogue systems, these changes are primarily reflected in the addition or alteration of *person*, *time*, *location*, and *topic*. Given l sessions denoted as $S = \{s_1, \dots, s_l\}$, natural dialogue boundaries are formed during session transitions (e.g. from s_i to s_{i+1}). In session s_i , we simulate the cortex to extract boundaries. The content between two boundaries forms what we refer to as boundary-guided memory.

To facilitate more intuitive indexing and semantic information, we establish nodes based on the various elements (i.e. person, time, location, and topic) within this memory and create an indexing interface. Building upon this semantic information, we further connect nodes of different elements within the same segment to form a hyperedge, representing the boundary-guided memory. Specifically, we construct a boundary extraction prompt P_{BE} and invoke LLMs (ϕ) to obtain the memory B_i , which contains the list of element nodes N_i and hyperedges H_i for s_i , as shown below:

$$\phi(s_i|P_{BE}) \rightarrow B_i = (N_i, H_i). \quad (1)$$

The list N_i consists of person nodes P_i , time nodes T_i , location nodes L_i , and topic nodes C_i . We can define any node n as follows:

$$\text{node} : n = (\text{name}, \text{mentions}, [\text{optional}] \text{granularity}), \quad (2)$$

where *name* serves as a unique identifier in s_i and *mentions* refers to the list of mentions that appears in s_i . When $n \in T_i$, additional granularity regarding time is extracted to present its finest division, thus preventing the model from generating incorrect time due to

hallucination. The hyperedge h^j in H_i is mainly composed of a subset $(\tilde{P}^j, \tilde{T}^j, \tilde{L}^j, \tilde{C}^j)$ of nodes from all the elements associated with the corresponding segment (P_i, T_i, L_i, C_i) , as shown below:

$$\text{hyperedge} : h^j = (\tilde{P}^j, \tilde{T}^j, \tilde{L}^j, \tilde{C}^j, d^j, r^j), \quad (3)$$

We utilize the description d_i to retain semantics from several turns within each hyperedge. Furthermore, we constrain the model to analyze the reasons r_i for segmenting the boundaries, thereby enhancing the accuracy of extracting key elements that undergo change. Specifically, r_i is selected from $\{\text{change person, change time, change location, topic shift, and explicit marker}\}$.

3.2.2 Memory Construction. Leveraging the hippocampus’s role in the continuous integration of new information into long-term memory, we utilize hippocampal–cortical interactions to guide the sustained encoding of boundary memories B_i and enhance the enduring retention of details in s_i . This process ($M \rightleftharpoons B_i, i = 1 \dots l$) enables establishing boundary-guided long-term memory M .

The core of this process lies in the effective integration of new and existing boundary memories. For each node, we merge relevant mentions based on unique identifiers. During this process, timestamps are converted into ISO 8601 format according to the time granularity. For instance, the term *yesterday* is transformed into the ‘%Y-%m-%d’ format based on the session time or a specific time in the session. Additionally, we compute the salience score of each node to reflect its importance within the overall dialogue. The salience score takes into account the following three dimensions:

Frequency: This measures the number of times the node appears in long-term memory, thus determining its significance.

Centrality: This analyzes the degree of the node within the hypergraph structure, as more central nodes are typically more important.

Diversity: This assesses the co-occurrence of the node with others, reflecting how common it is across different contexts.

For hyperedges, we determine whether to merge them by calculating the field-aware Jaccard Score (J) between pairs of hyperedges (h^i and h^j). First, we compute a node set for each hyperedge $N_{(h^i)}$ and then compute the score for any two hyperedges as follows:

$$J(h^i, h^j) = |N_{(h^i)} \cap N_{(h^j)}| / |N_{(h^i)} \cup N_{(h^j)}|. \quad (4)$$

We proceed to merge hyperedges recursively until no pair of hyperedges has a Jaccard score exceeding 0.8. This strategy not only enhances the storage efficiency of long-term memory by avoiding redundancy but also ensures the accuracy and relevance of the information in boundary-guided memories.

We further analyze all hyperedges and utilize the theme clustering prompt P_{TC} to identify common topics C_{common} that frequently appear in long-term memory, as well as rare topics C_{rare} that are mentioned less often. This information is crucial for developing subsequent retrieval strategies. To accommodate different categories of user queries, we should deploy varied recall strategies based on the frequency of mentions. For example, we should assess the quantity of recalls for common topics while emphasizing specific key pieces of memory for rare topics. The boundary-guided long-term memory can finally be represented as:

$$M = \{N, H, C_{common}, C_{rare}\}. \quad (5)$$

Thus, we have established a boundary-guided long-term memory and provided four explicit element node indices. By incorporating the structured information from hyperedges, the system effectively delivers both semantic and graph-related information.

3.3 Query Adaptive Retrieval

3.3.1 Retrieval Plan Generation. To address the diverse categories of queries, the dialogue system needs to analyze these queries and provides targeted retrieval plans that clarify ‘*what*’ the system needs to retrieve and ‘*how much*’ historical content is required. Specifically, we employ a query analysis prompt P_Q to process the queries, resulting in retrieval plans that encompass the following key information: the predicted type of query, the element and name to be queried, and the element priority.

Regarding query types, to ensure the applicability of the system, we do not directly predict the categories of queries delineated in specific datasets (such as LOCOMO). Instead, we categorize queries into three types based on the retrieval process: Recall Priority, Precision Priority, and Judgment. Recall priority signifies that it is essential to retrieve as many hyperedges in M related to the query as possible for an effective response. Precision priority, on the other hand, indicates that the most relevant individual or a few hyperedges are required to answer the question, rendering other relevant memories unnecessary in this context. Judgment query only requires identifying any a few of relevant pieces of memory to answer, as excessive retrieval may lead to redundancy.

Regarding query elements, the retrieval plan specifies the relevant interfaces that need to be queried for the query (*i.e.* elements person, time, location, and topic). Additionally, P_Q also asks to extract the *names* that provide information under these interfaces.

Regarding element priority, we believe that the order of priority for query interfaces is crucial for the adaptive retrieval of each query. For example, in the query ‘When did Caroline have lunch at KFC?’, the priority of time is significantly higher than person and location. Therefore, the retrieval plan includes the ranking of different elements (p) to enhance the retrieval process.

From a structural perspective, based on the query elements and their associated names, HingeMem identifies the corresponding nodes ($\hat{N}$). This enables system to obtain hyperedges that encompass these nodes. From a semantic perspective, HingeMem matches the query with the similarity of the embeddings of the hyperedges’ descriptions. Consequently, HingeMem derives a set of candidate hyperedges ($\hat{H}$) from the boundary-guided long-term memory M , with each hyperedge’s initial score (ξ) being its similarity value.

3.3.2 Hyperedge Rerank. A straightforward selection from the similarity between query (q) and hyperedges ($\hat{H}$) can lead to suboptimal outcomes, particularly in scenarios with diverse query types and varying relevance of retrieved memories. To enhance the retrieval process, we propose a reranking mechanism by incorporating the relationships among hyperedges within long-term memory. This mechanism prioritizes hyperedges based on the salience values of the involved nodes, ensuring that more pertinent memories are emphasized. Furthermore, we propose to add a penalty term to prevent less frequent memories from being overshadowed in the long-term memory. This penalty term considers the proximity of the hyperedges to identified rare and common topics, allowing for

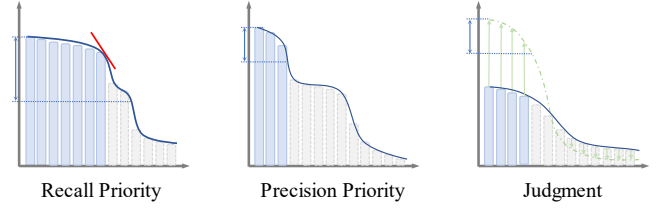


Figure 3: Diagram of adaptive stop for different query types.

a nuanced distinction between widely referenced and less frequent yet relevant information.

The computation of impact (Ω_S) of salience values involves a weighted sum of the elements based on their priority p as obtained in the retrieval plan. The element scores are obtained by averaging the salience values of all nodes in the corresponding node list (such as $\hat{P}^i$). For the topic penalty term (Ω_T), we calculate the distance from the current query to rare terms and subtract the distance to common terms. The definitions of rare terms and common terms are derived from the respective rare and common topics. To mitigate the issue of overlapping topics between the C_{common} and C_{rare} , we first calculate the feature subspaces for both rare and common topics. Subsequently, we remove any overlapping subspaces and assess the cross-similarity to generate weights accordingly. After computing the similarity between the query and each topic in C_{common} and C_{rare} , we finally apply a weighted softmax aggregation to derive the rare and common terms for the specific query. The updated score $\hat{\xi}^i$ for each hyperedge $\hat{h}^i$ with its initial score ξ^i according to query q is computed as follows:

$$\hat{\xi}^i = \xi^i + \Omega_S(\hat{N}_{(\hat{h}^i)}|p^i) + \Omega_T(q|C_{common}, C_{rare}). \quad (6)$$

This strategy to reranking ensures that the final selection of hyperedges is not only more relevant to the current query but also presents a balanced view of existing personal knowledge in the long-term memory by considering both common and rare details.

3.3.3 Adaptive Stop. Unlike existing methods, we do not simply select the Top-k hyperedges as context. Instead, we adopt an adaptive approach to determine the appropriate number of memory contents based on the query type. This strategy not only reduces overall unnecessary token costs but also mitigates the risk of introducing excessive distracting information, which could lead to incorrect answers. For details on when to stop selecting relevant hyperedges, refer to Figure 3. Here, we first sort the updated score $\hat{\xi}$.

For **Recall Priority** query, we detect inflection points in the score changes. Specifically, we select hyperedges $\hat{h}^i$ before the index $\min\{i : \hat{\xi}^{i+1} - \hat{\xi}^i < \lambda_{knee} \wedge \hat{\xi}^i > \max(\hat{\xi})/2\}$ ². This enables us to focus on a sufficient number of relevant memories while avoiding the noise that may arise from lower-quality hyperedges.

For **Precision Priority** query, we prioritize the most confident data segments. Specifically, we dynamically scale the acceptable confidence to select hyperedges where the score ($\hat{\xi}^i$) exceeds 80% of the maximum score $\max(\hat{\xi})$. This ensures that the retrieved hyperedges are not only relevant but also reliable, thereby enhancing the overall response accuracy.

²In practice, we set λ_{knee} to 0.1.

Table 1: Results on the LOCOMO dataset across five question categories. We report F1 scores (F_1), BLEU-1 scores (B_1), and LLM-as-a-Judge scores (J). The higher values indicate better performances. $\dagger$ denotes scores reported by the A-Mem paper. ‘Cat.’ indicates whether to apply the different QA format for each category. Best results are shown in Bold.

Method	Cat.	Single-Hop		Multi-Hop		Temporal		Open-Domain		Adversarial		Overall		
		F_1	J	F_1	J	F_1	J	F_1	J	F_1	J	F_1	J	B_1
LOCOMO	✓	12.7	16.5	19.7	20.9	10.4	11.5	20.1	30.2	66.8	90.1	25.8	33.5	0.132
RAG (Top-5)	✓	32.4	47.4	26.9	38.6	22.5	18.0	21.4	35.4	90.6	90.3	42.6	50.5	0.293
RAG (Top-10)	✓	36.4	50.7	29.6	38.6	27.7	22.7	21.2	37.5	86.8	86.5	44.6	51.9	0.306
RAG (Top-20)	✓	37.7	52.5	34.9	48.9	30.0	22.7	21.9	41.6	83.2	82.9	45.5	53.5	0.315
ReadAgent †	✓	6.6	-	2.5	-	5.3	-	10.1	-	5.4	-	5.7	-	0.109
MemGPT †	✓	10.4	-	4.2	-	13.4	-	9.5	-	31.5	-	14.6	-	0.125
A-Mem †	✓	18.2	-	24.3	-	16.4	-	23.6	-	46.0	-	25.2	-	0.208
Memorybank	✗	6.8	63.0	9.1	33.6	4.6	34.8	5.6	35.2	0.0	59.1	5.2	52.0	0.031
+ Cat. format	✓	39.9	59.2	23.3	28.7	23.3	31.7	20.9	44.7	86.3	86.3	44.4	55.8	0.303
LangMem	✗	54.7	70.1	47.4	58.5	48.6	51.0	27.8	38.3	17.3	81.1	42.9	62.5	0.307
+ Cat. format	✓	54.1	70.1	45.6	58.5	46.8	51.7	21.5	44.1	87.5	87.0	57.6	64.3	0.392
HippoRAG2	✗	54.4	78.5	35.4	52.8	55.5	61.0	23.4	35.2	4.3	69.5	39.1	68.5	0.289
+ Cat. format	✓	59.2	75.5	38.0	46.4	44.9	66.6	21.2	35.4	87.7	87.2	58.4	70.6	0.396
Mem0	✗	45.1	56.7	42.7	48.2	49.7	50.1	27.7	47.2	6.5	56.6	36.0	53.7	0.254
+ Cat. format	✓	44.0	59.0	38.6	47.8	45.6	47.6	20.8	42.0	84.3	72.7	51.4	59.6	0.351
Mem0 _g	✗	45.3	55.1	40.4	48.5	47.9	52.0	28.4	41.0	6.7	54.1	35.5	51.7	0.258
+ Cat. format	✓	43.4	62.0	38.3	45.7	44.4	48.5	23.4	44.1	82.7	68.5	50.7	59.7	0.348
Zep	✗	20.6	70.0	22.0	52.4	21.7	52.6	6.6	40.4	2.9	54.1	16.3	59.6	0.098
+ Cat. format	✓	59.4	76.7	41.1	52.8	50.0	66.9	22.4	40.0	74.7	74.4	56.9	69.4	0.406
HingeMem	✗	61.1	78.8	53.6	62.8	57.4	66.9	30.7	46.4	87.4	87.8	63.9	75.1	0.404

Table 2: Statistics of the LOCOMO dataset.

Conversation Statistics	Counts	Question Statistics	Counts
Total Conversations	10	Single-hop Questions	841
Avg. Sessions in a conversation	27.2	Multi-hop Questions	282
Avg. Turns in a conversation	294.1	Temporal Questions	321
Avg. Tokens in a conversation	15965.8	Open-domain Questions	96
Avg. Tokens in a session	587.0	Adversarial Questions	446
Avg. Tokens in a Turn	27.1	Total	1986

For **Judgment** query, we aim to select data that has a significant impact on the query. We initially apply a softmax function to scale all candidate hyperedges. In this way, even if many hyperedges that fulfill the query have relatively low scores, we can still select any one of them after rescaling. Subsequently, we choose hyperedges with softmax scores greater than 80% of the maximum softmax score. This strategy emphasizes the hyperedges that are likely to provide the most valuable information for making judgments.

Ultimately, the selected hyperedges serve as the context for generating the final answer to the current query. Notably, HingeMem does not require the use of a specific question template for different categories of queries, such as those provided by LOCOMO³.

4 Experiment

4.1 Experimental Settings

4.1.1 Datasets. We evaluate the long-term conversational memory in dialogue systems using the LOCOMO [29] dataset, which contains substantially longer dialogues than previous datasets, such

³For example, Line 243-259 in *task_eval/gpt_utils.py* in its official repository.

as MSC [48] and Conversation Chronicles [21]. As summarized in Table 2, LOCOMO comprises multi-session conversations averaging 15K tokens and spanning up to 27 sessions⁴, making it well-suited for assessing systems’ ability to construct long-term memory and maintain cross-session consistency over extended interactions. Each dialogue involves two participants discussing daily experiences or past events over a few months. Following each conversation, the dataset provides ~200 questions with corresponding ground-truth answers, covering five categories: single-hop, multi-hop, temporal, open-domain, and adversarial.

4.1.2 Metrics. Following the previous research in conversational AI [17, 38], we report the lexical metrics F1 Score (F_1) and BLEU-1 (B_1). While answers with substantial lexical overlap may still contain critical factual errors. To mitigate this limitation, we also employ an LLM-as-a-Judge (J) as a complementary evaluation metric. The judge model analyzes the question, ground-truth answer, and the system’s answer, delivering a more nuanced evaluation that better aligns with human judgment. To ensure a fair comparison, we follow the LOCOMO evaluation protocol: computing F_1 scores separately for each question category with different rules and then aggregating to obtain the overall F_1 scores. For the LLM-as-a-Judge, we apply the same prompt template as Mem0 [7].

4.1.3 Baselines. We compare the proposed HingeMem against two types of baselines distinguished by their memory structures: (1) **Only Semantics:** LOCOMO [29], Retrieval-Augmented Generation

⁴Data is sourced from the LOCOMO Github repository and differs from the version reported in the original paper. The authors provide a more representative subset to enable efficient validation for the community. In contrast, LongMemEval [46] contains only 500 questions, each tied to a single dialogue. This makes evaluation inefficient and environmentally unfriendly. Therefore, we primarily use LOCOMO in this paper.

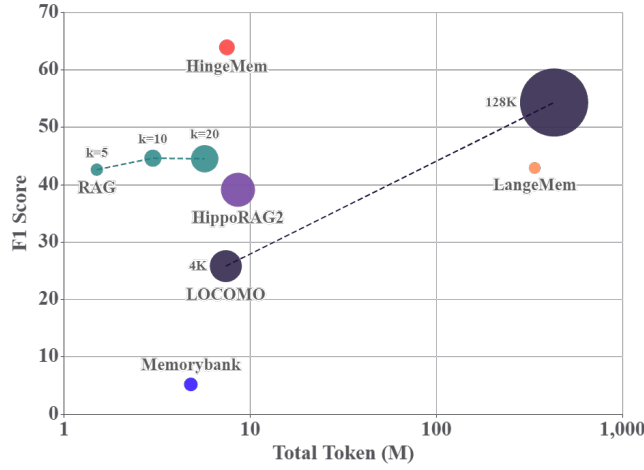


Figure 4: A comparative analysis of efficiency across different methods. For LOCOMO, we employ two configurations with context lengths of 4K and 128K. For RAG, we establish several parameter settings ranging from Top-5 to Top-20. The horizontal axis represents the total number of tokens consumed during the entire memory construction (if applicable) and the question-answering process. The circle size corresponds to the total tokens expended for all the users’ queries.

(RAG) [29], ReadAgent [24], MemGPT [32], A-Mem [49], Memorybank [56], and Mem0 [7]; (2) **Semantics and Graphs**: LangMem, HippoRAG2 [18, 19], Mem0 with graph memory (Mem_g) [7], and Zep [34]. For a fair comparison, we conduct experiments using their open-source implementation under identical experimental settings. Unless otherwise specified, all experiments are conducted with GPT-4o via the official structured output API. We use text-embedding-3-small to achieve all the text embeddings.

4.2 Main Results

4.2.1 Superiority. Table 1 demonstrates that HingeMem achieves optimal performance across various categories of queries. Notably, it shows an over 5% improvement in overall metrics and an impressive enhancement exceeding 10% on multi-hop questions. This significant progress is primarily attributed to our proposed boundary-guided memory, which effectively captures more memory details from chaotic dialogues. Although RAG (Top-5) achieves the best results on adversarial questions, HingeMem remains competitive in performance. RAG provides limited information by selecting only the top five pieces of content. Furthermore, it often retrieves irrelevant information in extensive dialogues, making it hard to fall into hallucinations when confronted with LLM’s powerful contextual comprehension ability and the input of short contexts.

Additionally, HingeMem, with the proposed adaptive retrieval, excels without the need to know the question type. In contrast, existing methods experience substantial performance declines when not informed of the category of questions. Memorybank and Zep show the most significant drops, ranging from 30% to 40%. Overall, HingeMem reveals its superiority across multiple critical metrics,

Table 3: Ablation study. TM and BM represent the common Textual Memory and our Boundary Guided Memory. ② only uses the hyperedges’ description for retrieval. NI means Node Indexing, which utilizes the structural information in hyperedges. HR and AS represent the proposed Hyperedge Rerank and Adaptive Stop.

# Num	Method	All	Each Category					
①	RAG+TM	44.6	36.4	29.6	27.7	21.2	86.8	
②	RAG+BM	57.4	54.5	39.3	50.1	24.3	86.8	
③	RAG+BM+NI	58.1	56.1	43.6	51.2	27.7	82.7	
④	③+HR	61.2	59.5	47.4	52.9	28.2	86.1	
⑤	③+HR+AS	63.9	61.1	53.6	57.4	30.7	87.4	

confirming its efficacy in dealing with complex conversations for practical long-term memory.

4.2.2 Efficiency. We conduct experiments to assess the impact of different methods and settings on efficiency. For Mem0 and Zep, we are unable to obtain an accurate token cost as we use their official services. Therefore, we exclude them in this comparison. The results are presented in Figure 4. While the memory construction process can be completed offline, the computation cost during the inference phase in response to user requests is of greater significance. We report the total token consumption during the experiments as well as the token cost associated with the question-answering phase.

The results indicate that increasing the context size (to 128K) allows for the transmission of more personalized information to the dialogue system, thereby significantly enhancing performance. However, this improvement comes with an unacceptable computational cost, reaching up to 0.4 billion tokens. Vanilla RAG alleviates the high computational demand by retrieving relevant content from historical dialogue to be incorporated into the context, while maintaining moderate performance. Nevertheless, it fails to effectively build long-term personal memory, as it requires searching through the entire history, which increases the inference overhead. Furthermore, vanilla RAG complicates the maintenance of historical data, making it more challenging to quickly and accurately locate the information needed by the user.

As Table 1 illustrates that constructing long-term memory contributes to improved system performance. Figure 4 further reveals that leveraging long-term memory can significantly reduce computational costs during the question-answering phase. Notably, HippoRAG2 stands as an exception, sustaining computational costs similar to those of RAG and LOCOMO. This is attributed to its use of unconstrained OpenIE to organize complex graph memories, necessitating additional attention during retrieval. It can also be observed that the memory construction of LangMem appears to be excessively inefficient compared to other methods. In contrast, our HingeMem demonstrates superior performance with total token and inference token costs comparable to existing approaches.

4.3 Ablation Study

4.3.1 Effectiveness of Boundary Guided Memory. We conduct experiments using vanilla RAG on both plain text-based memory and the proposed boundary guided memory to investigate the superiority of the boundary-guided memory. The results are illustrated

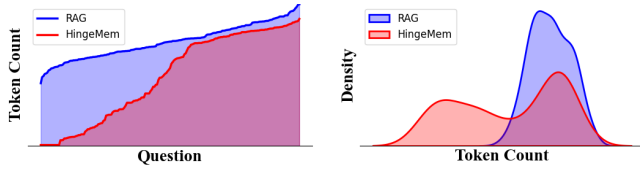


Figure 5: Analysis of the token count and its distribution of retrieval results for the question in sample 'conv-26' in LOCOMO, comparing RAG and HingeMem. HingeMem adaptively sets the retrieval capacity for different categories of questions, thereby reducing unnecessary noise and ultimately enhancing performance and efficiency.

in Table 3. Comparing ① and ②, we find that boundary-guided memory significantly enhances overall performance, resulting in an improvement of over 10%. This benefits from its ability to capture more detailed segments in the dialogues. However, because it does not intervene in the retrieval process, only using boundary-guided memory cannot reduce the noise memory present in the final selected context, resulting in no improvement for adversarial questions. Building on semantic retrieval, we further introduce node indexing based on information related to elements in the query to incorporate additional structural information. The results in ③ indicate that the integration of semantics and structure leads to further performance improvements. Particularly for multi-hop questions, we observe a near 4% improvement.

4.3.2 Effectiveness of Query Adaptive Retrieval. We conduct a detailed evaluation of the role of the proposed adaptive retrieval from both performance and efficiency perspectives. As shown in Table 3, ④ indicates a slight improvement following the introduction of hyperedge rerank. This enhancement arises from the strategy's consideration of the relationship between query and long-term memory, as well as the influences among different pieces of memory within the long-term memory. A comparison between ④ and ⑤ demonstrates that the dialogue system achieves optimal performance across various categories of questions by utilizing our adaptive stop strategy.

Query Adaptive Retrieval not only ensures the response accuracy but also reduces unnecessary token costs. We compare the context lengths retrieved by RAG and HingeMem for the same set of questions, as shown in Figure 5. The context length obtained by Top-K fluctuates within a certain range, making it vulnerable to noise interference and the loss of related information. In HingeMem, adaptive retrieval enables the specification of targeted retrieval plans for each query, thereby reducing overall token cost while ensuring that the retrieval results are informative and free of redundancy.

4.3.3 Robustness of Different Model Scales. We conduct evaluations on the advanced Qwen3-Series [51] model of different scales to validate the generalization ability of existing methods. The results in Figure 6 show that HingeMem maintains stable and superior performance across all scales, from the 0.6B model to the flagship model, demonstrating strong adaptability to changes in model size. Specifically, as the model scale increases, the performance curves exhibit a consistently rising trend, whereas the other baselines display

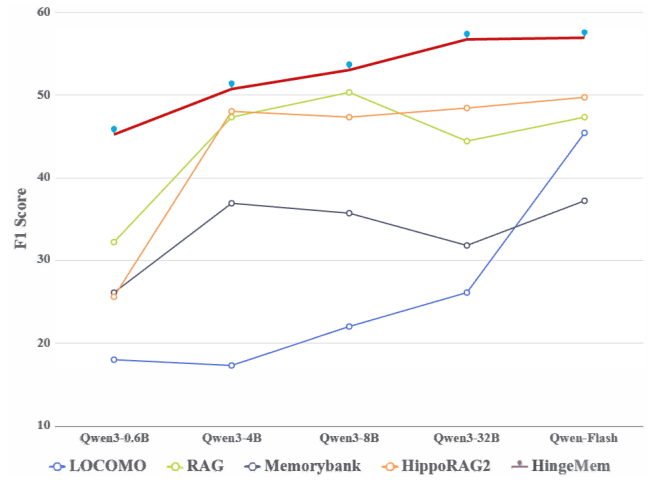


Figure 6: The performance comparison of different model scales in the Qwen-Series models. HingeMem delivers consistent and optimal results from the 0.6B model to the flagship model. This demonstrates the adaptability of our approach across various potential application scenarios.

fluctuations or declines at certain scales. This suggests that existing methods are relatively dependent on powerful base LLMs. Notably, in small-scale models, the vanilla RAG setup already achieves performance that surpasses other methods with explicitly constructed memory. These findings indicate that our proposed HingeMem generalizes robustly across computing powers and remains effective not only on web servers but also on resource-constrained mobile edge devices, enabling adaptive inference from cloud to on-device deployment with broad application potential.

5 Conclusion

In this work, we introduce HingeMem, inspired by hippocampal-cortical interactions and event segmentation theory. HingeMem extracts the boundary memory by locating the changes in key elements of events: person, time, location, and topic. Then it constructs the structured hyperedges from these elements and updates the boundary guided long-term memory. By incorporating adaptive retrieval strategies, HingeMem addresses the critical challenges of efficient memory retrieval for queries of different categories. Extensive experiments demonstrate that HingeMem outperforms existing state-of-the-art methods, offering a 5% improvement on LOCOMO while maintaining scalability across various scales of LLMs. HingeMem not only provides a practical solution to the limitations of existing memory mechanisms but also opens new avenues for research in memory design and efficient knowledge retrieval for conversational AI and intelligent Web applications.

Acknowledgements

This work was supported by the National Natural Science Foundation of China (U23B2057), and Shanghai Pilot Program for Basic Research (22TQ1400300).

References

- [1] Layla El Asri, Hannes Schulz, Shikhar Sharma, Jeremie Zumer, Justin Harris, Emery Fine, Rahul Mehrotra, and Kaheer Suleman. 2017. Frames: a corpus for adding memory to goal-oriented dialogue systems. In *SIGDIAL Conference*. Association for Computational Linguistics, 207–219.
- [2] Sanghwan Bae, Dong-Hyun Kwak, Soyoung Kang, Min Young Lee, Sungdong Kim, Yui Jeong, Hyeri Kim, Sang-Woo Lee, Woo-Myoung Park, and Nako Sung. 2022. Keep Me Updated! Memory Management in Long-term Conversations. In *EMNLP (Findings)*. Association for Computational Linguistics, 3769–3787.
- [3] Christopher Baldassano, Janice Chen, Asieh Zadbood, Jonathan W Pillow, Uri Hasson, and Kenneth A Norman. 2017. Discovering event structure in continuous narrative perception and memory. *Neuron* 95, 3 (2017), 709–721.
- [4] Alexander J Barnett, Mitchell Nguyen, James Spargo, Reesha Yadav, Brendan I Cohn-Sheehy, and Charan Ranganath. 2024. Hippocampal-cortical interactions during event boundaries support retention of complex narrative events. *Neuron* 112, 2 (2024), 319–330.
- [5] Yuxi Bi, Yunfan Gao, and Haofen Wang. 2025. StePO-Rec: Towards Personalized Outfit Styling Assistant via Knowledge-Guided Multi-Step Reasoning. *CoRR* abs/2504.09915 (2025).
- [6] Irving Biederman. 1987. Recognition-by-components: a theory of human image understanding. *Psychological review* 94, 2 (1987), 115.
- [7] Prateek Chhikara, Dev Khant, Saket Arya, Aranjeet Singh, and Deshray Yadav. 2025. Mem0: Building Production-Ready AI Agents with Scalable Long-Term Memory. *CoRR* abs/2504.19413 (2025).
- [8] Finetune Corp. 2024. Memory: The Open Source Memory Layer For Autonomous Agents. <https://github.com/kingjuli08238/memory>.
- [9] Yiming Du, Wenyu Huang, Danna Zheng, Zhaowei Wang, Sébastien Montella, Mirella Lapata, Kam-Fai Wong, and Jeff Z. Pan. 2025. Rethinking Memory in AI: Taxonomy, Operations, Topics, and Future Directions. *CoRR* abs/2505.00675 (2025).
- [10] Yiming Du, Hongru Wang, Zhengyi Zhao, Bin Liang, Baojun Wang, Wanjuan Zhong, Zezhong Wang, and Kam-Fai Wong. 2024. PerLQA: A Personal Long-Term Memory Dataset for Memory Classification, Retrieval, and Synthesis in Question Answering. *CoRR* abs/2402.16288 (2024).
- [11] Darren Edge, Ha Trinh, Newman Cheng, Joshua Bradley, Alex Chao, Apurva Mody, Steven Truitt, and Jonathan Larson. 2024. From Local to Global: A Graph RAG Approach to Query-Focused Summarization. *CoRR* abs/2404.16130 (2024).
- [12] Institute for Basic Science. 2023. AI's memory-forming mechanism found to be strikingly similar to that of the brain. *ScienceDaily* (2023).
- [13] Nicholas T Franklin, Kenneth A Norman, Charan Ranganath, Jeffrey M Zacks, and Samuel J Gershman. 2020. Structured Event Memory: A neuro-symbolic model of event cognition. *Psychological review* 127, 3 (2020), 327.
- [14] Yunfan Gao, Tao Sheng, Youlin Xiang, Yun Xiong, Haofen Wang, and Jiawei Zhang. 2023. Chat-REC: Towards Interactive and Explainable LLMs-Augmented Recommender System. *CoRR* abs/2303.14524 (2023).
- [15] Yubin Ge, Salvatore Romeo, Jason Cai, Raphael Shu, Yassine Benajiba, Monica Sunkara, and Yi Zhang. 2025. TRem: Towards Neuro-Symbolic Temporal Reasoning for LLM-Agents with Memory in Multi-Session Dialogues. In *ACL (Findings)*. Association for Computational Linguistics, 18974–18988.
- [16] Linda Geerligs, Dora Gözikara, Djamar Oetinger, Karen L Campbell, Marcel van Gerven, and Umut Güçlü. 2022. A partially nested cortical hierarchy of neural states underlies event segmentation in the human brain. *elife* 11 (2022), e77430.
- [17] Gaurav Goswami. 2025. Dissecting the metrics: How different evaluation approaches yield diverse results for conversational ai. *Authorea Preprints* (2025). TechRiv:26407 <https://www.techrxiv.org/inst/26407>
- [18] Bernal Jimenez Gutierrez, Yiheng Shu, Yu Gu, Michihiro Yasunaga, and Yu Su. 2024. HippoRAG: Neurobiologically Inspired Long-Term Memory for Large Language Models. In *NeurIPS*.
- [19] Bernal Jiménez Gutiérrez, Yiheng Shu, Weijian Qi, Sizhe Zhou, and Yu Su. 2025. From RAG to Memory: Non-Parametric Continual Learning for Large Language Models. *CoRR* abs/2502.14802 (2025).
- [20] Junqing He, Liang Zhu, Rui Wang, Xi Wang, Gholamreza Haffari, and Jiaxing Zhang. 2025. MADial-Bench: Towards Real-world Evaluation of Memory-Augmented Dialogue Generation. In *NAACL (Long Papers)*. Association for Computational Linguistics, 9902–9921.
- [21] Jihyung Jang, Minseong Boo, and Hyoungun Kim. 2023. Conversation Chronicles: Towards Diverse Temporal and Relational Dynamics in Multi-Session Conversations. In *EMNLP*. Association for Computational Linguistics, 13584–13606.
- [22] Eunwon Kim, Chanho Park, and Buru Chang. 2025. SHARE: Shared Memory-Aware Open-Domain Long-Term Dialogue Dataset Constructed from Movie Script. In *ACL (1)*. Association for Computational Linguistics, 14474–14498.
- [23] Seo Hyun Kim, Keummin Ka, Yohan Jo, Seung-won Hwang, Dongha Lee, and Jinyoung Yeo. 2024. Ever-Evolving Memory by Blending and Refining the Past. *CoRR* abs/2403.04787 (2024).
- [24] Kuang-Huei Lee, Xinyun Chen, Hiroki Furuta, John F. Canny, and Ian Fischer. 2024. A Human-Inspired Reading Agent with Gist Memory of Very Long Contexts. In *ICML*. OpenReview.net.
- [25] Zhiyu Li, Shichao Song, Chenyang Xi, Hanyu Wang, Chen Tang, Simin Niu, Ding Chen, Jiawei Yang, Chunyu Li, Qingchen Yu, Jihao Zhao, Yezhaohui Wang, Peng Liu, Zehao Lin, Pengyuan Wang, Jiahao Huo, Tianyi Chen, Kai Chen, Kehang Li, Zhen Tao, Junpeng Ren, Huayi Lai, Hao Wu, Bo Tang, Zhenren Wang, Zhaoxin Fan, Ningyu Zhang, Linfeng Zhang, Junchi Yan, Mingchuan Yang, Tong Xu, Wei Xu, HuaJun Chen, Haofeng Wang, Hongkang Yang, Wentao Zhang, Zhi-Qin John Xu, Siheng Chen, and Feiyu Xiong. 2025. MemOS: A Memory OS for AI System. *CoRR* abs/2507.03724 (2025).
- [26] Lei Liu, Xiaoyan Yang, Yue Shen, Binbin Hu, Zhiqiang Zhang, Jinjie Gu, and Guannan Zhang. 2023. Think-in-Memory: Recalling and Post-thinking Enable LLMs with Long-Term Memory. *CoRR* abs/2311.08719 (2023).
- [27] Junru Lu, Siyu An, Mingbao Lin, Gabriele Pergola, Yulan He, Di Yin, Xing Sun, and Yunsheng Wu. 2023. MemoChat: Tuning LLMs to Use Memos for Consistent Long-Range Open-Domain Conversation. *CoRR* abs/2308.08239 (2023).
- [28] Qihong Lu, Uri Hasson, and Kenneth A Norman. 2022. A neural network model of when to retrieve and encode episodic memories. *elife* 11 (2022), e74445.
- [29] Adyasha Maharana, Dong-Ho Lee, Sergey Tulyakov, Mohit Bansal, Francesco Barbieri, and Yuwei Fang. 2024. Evaluating Very Long-Term Conversational Memory of LLM Agents. In *ACL (1)*. Association for Computational Linguistics, 13851–13870.
- [30] Jisoo Mok, Ik-hwan Kim, Sangkwon Park, and Sungroh Yoon. 2025. Exploring the Potential of LLMs as Personalized Assistants: Dataset, Evaluation, and Analysis. In *ACL (1)*. Association for Computational Linguistics, 10212–10239.
- [31] Kai Tzu-iunn Ong, Namyoung Kim, Minju Gwak, Hyungjoo Chae, Taeyoon Kwon, Yohan Jo, Seung-won Hwang, Dongha Lee, and Jinyoung Yeo. 2025. Towards Lifelong Dialogue Agents via Timeline-based Memory Management. In *NAACL (Long Papers)*. Association for Computational Linguistics, 8631–8661.
- [32] Charles Packer, Vivian Fang, Shishir G. Patil, Kevin Lin, Sarah Wooders, and Joseph E. Gonzalez. 2023. MemGPT: Towards LLMs as Operating Systems. *CoRR* abs/2310.08560 (2023).
- [33] Advait Paliwal. 2024. Reminisce: Memory for LLMs. <https://github.com/advaitpaliwal/reminisce>.
- [34] Preston Rasmussen, Pavlo Paliychuk, Travis Beauvais, Jack Ryan, and Daniel Chalef. 2025. Zep: A Temporal Knowledge Graph Architecture for Agent Memory. *CoRR* abs/2501.13956 (2025).
- [35] Zachariah M Reagh, Angelique I Delarazan, Alexander Garber, and Charan Ranganath. 2020. Aging alters neural activity at event boundaries in the hippocampus and Posterior Medial network. *Nature communications* 11, 1 (2020), 3980.
- [36] Alireza Salemi, Sheshera Mysore, Michael Bendersky, and Hamed Zamani. 2024. LaMP: When Large Language Models Meet Personalization. In *ACL (1)*. Association for Computational Linguistics, 7370–7392.
- [37] Gregor Sieber and Brigitte Krenn. 2010. Towards an Episodic Memory for Companion Dialogue. In *IVA (Lecture Notes in Computer Science, Vol. 6356)*. Springer, 322–328.
- [38] Arpita Soni, Rajeev Arora, Anoop Kumar, and Dheerendra Panwar. 2024. Evaluating Domain Coverage in Low-Resource Generative Chatbots: A Comparative Study of Open-Domain and Closed-Domain Approaches Using BLEU Scores. In *2024 International Conference on Electrical Electronics and Computing Technologies (ICEECT)*, Vol. 1. 1–6. doi:10.1109/ICEECT61758.2024.10738994
- [39] Hao Sun, Hengyi Cai, Bo Wang, Yingyan Hou, Xiaochi Wei, Shuaiqiang Wang, Yan Zhang, and Dawei Yin. 2024. Towards Verifiable Text Generation with Evolving Memory and Self-Reflection. In *EMNLP*. Association for Computational Linguistics, 8211–8227.
- [40] Jingwei Sun, Zhixu Du, and Yiran Chen. 2024. Knowledge Graph Tuning: Real-time Large Language Model Personalization based on Human Feedback. *CoRR* abs/2405.19686 (2024).
- [41] Wang-Chiew Tan, Jane Dwivedi-Yu, Yuliang Li, Lambert Mathias, Marzieh Saeidi, Jing Nathan Yan, and Alon Y. Halevy. 2023. TimelineQA: A Benchmark for Question Answering over Timelines. In *ACL (Findings)*. Association for Computational Linguistics, 77–91.
- [42] Zhen Tan, Jun Yan, I-Hung Hsu, Rujun Han, Zifeng Wang, Long T. Le, Yiwen Song, Yanfei Chen, Hamid Palangi, George Lee, Anand Rajan Iyer, Tianlong Chen, Huan Liu, Chen-Yu Lee, and Tomas Pfister. 2025. In Prospect and Retrospect: Reflective Memory Management for Long-term Personalized Dialogue Agents. In *ACL (1)*. Association for Computational Linguistics, 8416–8439.
- [43] Bing Wang, Xinnian Liang, Jian Yang, Hui Huang, Shuangzhi Wu, Peihao Wu, Lu Lu, Zejun Ma, and Zhoujun Li. 2025. SCM: Enhancing Large Language Model with Self-Controlled Memory Framework. In *DASFAA*. <https://arxiv.org/abs/2304.13343>
- [44] Qingyue Wang, Yanhe Fu, Yanan Cao, Shuai Wang, Zhiliang Tian, and Liang Ding. 2025. Recursively summarizing enables long-term dialogue memory in large language models. *Neurocomputing* 639 (2025), 130193.
- [45] Zheng Wang, Zhongyang Li, Zeren Jiang, Dandan Tu, and Wei Shi. 2024. Crafting Personalized Agents through Retrieval-Augmented Generation on Editable Memory Graphs. In *EMNLP*. Association for Computational Linguistics, 4891–4906.
- [46] Di Wu, Hongwei Wang, Wenhao Yu, Yuwei Zhang, Kai-Wei Chang, and Dong Yu. 2025. LongMemEval: Benchmarking Chat Assistants on Long-Term Interactive Memory. In *ICLR*. OpenReview.net.

- [47] Yaxiong Wu, Sheng Liang, Chen Zhang, Yichao Wang, Yongyue Zhang, Huifeng Guo, Ruiming Tang, and Yong Liu. 2025. From Human Memory to AI Memory: A Survey on Memory Mechanisms in the Era of LLMs. *CoRR* abs/2504.15965 (2025).
- [48] Jing Xu, Arthur Szlam, and Jason Weston. 2022. Beyond Goldfish Memory: Long-Term Open-Domain Conversation. In *ACL (1)*. Association for Computational Linguistics, 5180–5197.
- [49] Wujiang Xu, Zujie Liang, Kai Mei, Hang Gao, Juntao Tan, and Yongfeng Zhang. 2025. A-MEM: Agentic Memory for LLM Agents. *CoRR* abs/2502.12110 (2025).
- [50] Xinchao Xu, Zhibin Gou, Wenquan Wu, Zheng-Yu Niu, Hua Wu, Haifeng Wang, and Shihang Wang. 2022. Long Time No See! Open-Domain Conversation with Long-Term Persona Memory. In *ACL (Findings)*. Association for Computational Linguistics, 2639–2650.
- [51] An Yang, Anfeng Li, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, Chang Gao, Chengen Huang, Chenxu Lv, Chujie Zheng, Dayiheng Liu, Fan Zhou, Fei Huang, Feng Hu, Hao Ge, Haoran Wei, Huan Lin, Jialong Tang, Jian Yang, Jianhong Tu, Jianwei Zhang, Jian Yang, Jiaxi Yang, Jingren Zhou, Junyang Lin, Kai Dang, Keqin Bao, Kexin Yang, Le Yu, Lianghao Deng, Mei Li, Mingfeng Xue, Mingze Li, Pei Zhang, Peng Wang, Qin Zhu, Rui Men, Ruize Gao, Shixuan Liu, Shuang Luo, Tianhao Li, Tianyi Tang, Wenbiao Yin, Xingzhang Ren, Xinyu Wang, Xinyu Zhang, Xuancheng Ren, Yang Fan, Yang Su, Yichang Zhang, Yinger Zhang, Yu Wan, Yuqiong Liu, Zekun Wang, Zeyu Cui, Zhenru Zhang, Zhipeng Zhou, and Zihan Qiu. 2025. Qwen3 Technical Report. *CoRR* abs/2505.09388 (2025).
- [52] Jeffrey M Zacks and Khen M Swallow. 2007. Event segmentation. *Current directions in psychological science* 16, 2 (2007), 80–84.
- [53] Jeffrey M Zacks, Barbara Tversky, and Gowri Iyer. 2001. Perceiving, remembering, and communicating structure in events. *Journal of experimental psychology: General* 130, 1 (2001), 29.
- [54] Zheng Zhang, Minlie Huang, Zhongzhou Zhao, Feng Ji, Haiqing Chen, and Xiaoyan Zhu. 2019. Memory-Augmented Dialogue Management for Task-Oriented Dialogue Systems. *ACM Trans. Inf. Syst.* 37, 3 (2019), 34:1–34:30.
- [55] Jie Zheng, Andrea GP Schjetnan, Mar Yebra, Bernard A Gomes, Clayton P Mosher, Suneil K Kalia, Taufik A Valiante, Adam N Mamelak, Gabriel Kreiman, and Ueli Rutishauser. 2022. Neurons detect cognitive boundaries to structure episodic memories in humans. *Nature neuroscience* 25, 3 (2022), 358–368.
- [56] Wanjun Zhong, Lianghong Guo, Qiqi Gao, He Ye, and Yanlin Wang. 2024. MemoryBank: Enhancing Large Language Models with Long-Term Memory. In *AAAI*. AAAI Press, 19724–19731.
- [57] Yijie Zhong, Yunfan Gao, Xiaolian Zhang, and Haofen Wang. 2025. ODDA: An OODA-Driven Diverse Data Augmentation Framework for Low-Resource Relation Extraction. In *Findings of the Association for Computational Linguistics: ACL 2025*. 267–285.
- [58] Yijie Zhong, Feifan Wu, Mengying Guo, Xiaolian Zhang, Meng Wang, and Haofen Wang. 2025. Meta-PKE: Memory-Enhanced Task-Adaptive Personal Knowledge Extraction in Daily Life. *Inf. Process. Manag.* 62, 4 (2025), 104097.

A More Implementation Details

To enable fair comparison, we integrate all baselines into our project based on the following open-source codes. Then we use the same code for evaluation. Below, we focus on the most critical component: the prompt design.

- Locomo and RAG: <https://github.com/snap-research/LoCoMo>
- Memorybank: <https://github.com/zhongwanjun/MemoryBank-SiliconFriend>
- HippoRAG2: <https://github.com/OSU-NLP-Group/HippoRAG>
- LangMem and Mem0: <https://mem0.ai/research> and <https://github.com/mem0ai/mem0/tree/main/evaluation>
- Zep: <https://github.com/getzep/zep-papers> and <https://www.getzep.com/>

A.1 HingeMem

We present the Boundary Extract Prompt, Topic Clustering Prompt, and Query Analysis Prompt in our proposed HingeMem for reference. Here, we preserve the output schemas while omitting several additional principles in the prompt.

Prompt: Boundary Extraction

You are a "conversation segmentation and element extractor". Please perform the following tasks on the input complete conversation session (including datetime, turn id, and speaker name) and output only valid JSON.

Task Objectives

1. Identify event or memory boundaries: When there is an obvious change in **person/time/location/topic**, or a new **person/time/location/topic** appears, or when explicit transition words appear, start a new event; otherwise, merge it into the current memory data.
2. Extract relations and events in a unified form.

Fields: persons[], times[], locations[], topics[], description, boundaryreasons[], start turn, end turn

For relations: (optional) **Person - Person**: ... ; **Person - Time**: ... ; **Person - Location**: ...

For events: Fill in the corresponding fields according to the events involved and summarize the corresponding topics.

...

Output Specifications:

...

Output Schema

```
{ "persons": [ { "canonical_name": "<string>", "role_tags": ["<string>", ...], "mentions": [{"turn": <int>, "mention": "<string>"}, ...], ... },
  "times": [ { "timestamp": "<ISO8601>" (do not provide null, estimate based on the relative date if possible), "granularity": "second | minute | hour | day | week | month | year | approx", "mentions": [{"turn": <int>, "mention": "<string>"}, ...], ... },
  "locations": [ { "name": "<string>", "mentions": [{"turn": <int>, "mention": "<string>"}, ...], ... },
  "topics": [ { "label": "<string>", "mentions": [{"turn": <int>, "mention": "<string>"}, ...], ... },
  "boundary_memories": [ { "person_list": ["<canonical_name mentioned above>", ...], "time_list": ["<timestamp mentioned above>", ...], "location_list": ["<name mentioned above>", ...], "topic_list": ["<label mentioned above>", ...], "description": "<string>", "boundary": { "reasons": ["change_time | change_place | change_person | topic_shift | explicit_marker"], ... }, ... }, ... ]
***
{session data}
```

Prompt: Topic Clustering

Task: From INPUT_TOPICS (a list of historical event topics), select only from the input and return two lists.

Definitions:

- Common/Stable: widely documented, recurring across periods
- Rare/Uncommon: niche/localized, sparsely documented.

Rules:

- Pick 3–5 topics for each list; ...
- Concise noun phrases only; no dates or explanations.
- Do not invent; deduplicate; preserve original phrasing/case.
- Output valid JSON only (no extra text).

Return exactly this JSON shape:

```
{ "common_topics": [...], "rare_topics": [...]}
INPUT_TOPICS: {topics}
```

Prompt: Query Analysis

[Role]
 You are the "Query Analyzer". Please read the input query and output a structured search plan. The four elements you can use are: person, location, topic, time.

[Analyze Principles]
 - Question Type Definition (Choose One):
 - Recall-First: Questions that require coverage / enumeration / counting / timeline summarization. For example: - ...
 - Precision-First: Questions that seek the most relevant/single best piece of evidence. For example: - ...
 - Judgment-First: Yes / No / Existential decisions, or a balance of evidence. For example: - ...
 - If the person/location/topic/time is not clear, leave the corresponding fields blank and do not make them up.
 - "priority" only includes elements that exist in "constraints" [JSON Schema]
 {
 "query_type": "recall | precision | judgement",
 "constraints": {
 (optional) "person": ["str", ...],
 (optional) "location": ["str", ...],
 (optional) "topic": ["str", ...],
 (optional) "time": {
 "timestamp": "(ISO Time Format)",
 "granularity": "year | month | day | hour | minute | approx"
 }, ...],
 },
 "priority": [(sorted keys in constraints)] // List only the factors that you think are discriminatory, in order of priority
 }
 [Query]
 query
 [Output JSON]

A.2 LOCOMO Template

Below, we present the different templates used in LOCOMO for various question categories. For example, prompts for temporal questions require LLMs to return an explicit 'Date'. An inspection of existing methods' open-source code shows that they employ these category-specific prompts, which limits their practical applicability. In contrast, HingeMem adaptively handles various questions.

Prompt: Template for Temporal Questions

Based on the above context, write an answer in the form of a short phrase for the following question. Answer with exact words from the context whenever possible.
 Question: {Question} Use DATE of CONVERSATION to answer with an approximate date. Short answer:

Prompt: Template for Adversarial Questions

Based on the above context, answer the following question.
 Question: {Question} Select the correct answer: (a) {Adversarial Answer a} (b) {Adversarial Answer b}. Short answer:

Prompt: Template for Other Questions

Based on the above context, write an answer in the form of a short phrase for the following question. Answer with exact words from the context whenever possible.
 Question: {Question} Short answer:

B Details of Constructed Long-Term Memory

Table 4 presents the counts of element nodes and hyperedges in the long-term memory constructed by HingeMem on LOCOMO. The results show that HingeMem effectively captures fine-grained conversational details while avoiding the substantial redundancy and noise introduced by the continuous memory-writing strategies used in prior methods.

Table 4: Statistics of the constructed long-term memory.

Memory Statistics	Counts
Avg. Persons in a conversation	12.8
Avg. Times in a conversation	59.6
Avg. Locations in a conversation	34.6
Avg. Topics in a conversation	81.5
Avg. Hyperedges in a conversation	103.2

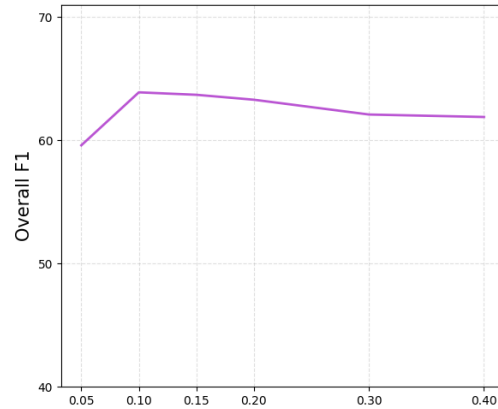
C Parameter Analysis**Figure 7: Performance of different values for λ_{knee}** **C.1 λ_{knee} for Recall-Priority Query**

Figure 7 shows the effect of the hyperparameter λ_{knee} on performance. If the inflection point threshold is set too small, the process halts prematurely, and valuable cues are lost. A moderate relaxation of this threshold yields the best performance. As λ_{knee} further increases, performance degrades but quickly reaches a floor. This is because (i) memory scores often do not exhibit abrupt, large-magnitude changes, rendering larger λ_{knee} has no practical effect, and (ii) the adaptive stopping mechanism for recall priority queries is also constrained by a maximum-score ratio, which further attenuates sensitivity to this hyperparameter.

C.2 Scale for Precision-Priority Query

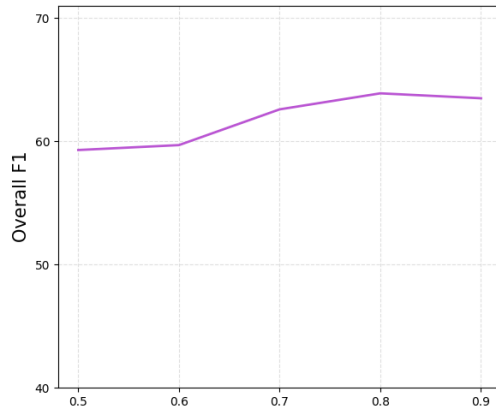


Figure 8: Impact of the scale for precision-priority query.

In practice, we scale the confidence to select hyperedges where the score exceeds 80% of the maximum score for precision-priority queries. Figure 8 shows the effect of the scale on performance. A small scale tends to admit numerous seemingly relevant yet inessential cues. To obtain more accurate and non-redundant cues, a larger scale is preferable. However, excessively large values (*i.e.* 0.9) do not yield additional performance gains and may narrow the scope of applicability. Accordingly, we set the scale to 0.8.

C.3 Scale for Judgment Query

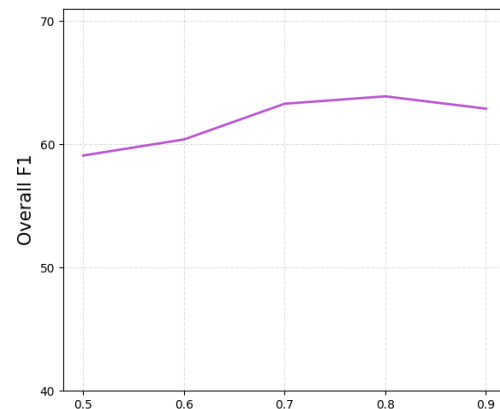


Figure 9: Impact of the scale for judgment query.

Figure 9 illustrates the effect of the scale applied to the relative maximum softmax score on judgment queries, exhibiting a trend similar to that observed for precision-priority queries. A small scale weakens the constraint, admitting spurious cues and degrading decision accuracy. Conversely, for judgment queries, an overly large scale has a more pronounced adverse impact on the final decision. Accordingly, we adopt a balanced setting of 0.8.